



Mastering GitHub Actions

MGA | 1 Day

Build, test, and ship faster with automated workflows. This one-day course is designed to help you put GitHub Actions to work across your real delivery pipeline, from your first workflow to production deployments. Through hands-on labs and real-world examples, you'll learn to create, secure, and manage workflows that streamline how your team builds and releases software.

Who should take this class?

This course is for developers, DevOps engineers, QA professionals, and technical team members who want to automate their build, test, and release workflows with GitHub Actions. It's equally valuable for tech leads and architects responsible for standardizing how their teams deliver software. Prior experience with GitHub and basic scripting (such as YAML or shell) is recommended, and a background in CI/CD or DevOps principles is helpful but not required.

Course content

This course progressively builds your fluency with GitHub Actions, from your first workflow to securing and scaling automated pipelines. You'll work hands-on in teams using a shared GitHub environment, configuring and running real workflows that mirror how production delivery actually works.

- 1. INTRODUCTION TO GITHUB ACTIONS**
 - GitHub Actions overview and core concepts
 - Events, workflows, jobs, steps, and actions
 - Workflow syntax and structure (YAML)
 - Creating and running your first workflow
 - GitHub-hosted, large, and self-hosted runners
 - Usage limits and guidelines
 - Migrating from Azure Pipelines
 - Hands-on lab
- 2. AUTOMATING BUILDS, TESTS, AND DEPLOYMENTS**
 - Configuring build and test pipelines (CI/CD)
 - Matrix builds and automated testing
 - Environment variables, secrets, and caching
 - Reusable workflows and composite actions
 - Securing workflows: GITHUB_TOKEN, least privilege, provenance
 - Deployment pipelines: Pages, cloud, and containers
 - Leveraging environments/deployment gating
 - Hands-on lab
- 3. ADVANCED TECHNIQUES AND BEST PRACTICES**
 - Advanced workflow design: conditions, expressions, and concurrency
 - Passing data between jobs with artifacts and outputs
 - Security and compliance: CODEOWNERS, pinned actions, GHAS, CodeQL, Dependabot
 - Secure cloud authentication with OpenID Connect (OIDC)
 - Reusing and standardizing workflows across an organization
 - Debugging, monitoring, and action usage metrics
 - Optimizing cost and performance (caching, canceling redundant runs)
 - Authoring workflows with VS Code and GitHub Copilot
 - Hands-on lab

Course Designer

This course was designed by Richard Hundhausen, a Professional Scrum Trainer, co-creator of the Nexus framework, and a longtime developer who helps delivery teams modernize how they build and ship software with DevOps, CI/CD, and AI-assisted engineering. To see other DevOps and AI courses, visit www.accentient.com.

